Web Scraping & Customer Sentiment Report

BestBuy Canada — Sony MDR-ZX110 Over-Ear Headphones | HanuAi Assessment

Platform:	Product SKU:	Reviews Scraped:	Analysis:
BestBuy Canada	10325154 (Sony MDR-ZX110)	179 Unique Records	VADER + Aspect Mapping

1. Overview & Setup

For this task, I collected and analyzed customer reviews from BestBuy Canada for the Sony MDR-ZX110 on-ear headphones. This model was chosen because it has a large volume of public reviews (over 11,000 on the site) and covers a broad range of user opinions. The goal was to build a reliable scraper that handles dynamic pagination and review filters, parse key review fields, perform sentiment classification, and map reviews into functional aspect tags.

2. Anti-Scraping Technical Solutions

2.1 Handling IP Blocking & Rate Limiting

- **Request Throttling & Jitter:** The scraper adds an exponential delay with random jitter (typically 1.5 to 3.0 seconds) between requests to avoid triggering basic rate-limit thresholds.
- **Residential Proxy Pools:** In production, routing requests through residential proxy providers (such as BrightData or Smartproxy) prevents IP range blacklisting targeting datacenter IP blocks.
- **Session Continuity:** Keeping a requests.Session() alive for consecutive pagination calls for the same SKU mimics a real user clicking through 'Show More'.

2.2 Managing CAPTCHA & Bot Detection

- **Direct JSON Endpoint Access:** BestBuy Canada's frontend queries an internal JSON API. Accessing this endpoint directly bypasses heavy client-side JavaScript execution, loads data faster, and avoids many front-end bot traps.
- **Browser Headers & User-Agent Rotation:** Every request sends standard desktop User-Agent headers, along with matching Referer, Accept-Language, and Origin headers.
- **Headless Browser Stealth:** When using Selenium/Playwright, stealth plugins override navigator.webdriver, emulate WebGL renderer strings, and simulate natural mouse trajectories.

2.3 Retry Logic & Connection Errors

Mechanism	Implementation Details	Purpose
Exponential Backoff	Wait = Base * (2 ^ attempt) + random(0.5, 1.5)s	Prevents overwhelming the server during temporary slowdowns
Retry Limits	Max 3 attempts per page before logging and moving on	Prevents infinite loops if a specific page or endpoint is down
Status Checks	Explicit checks for 429 (rate limit) and 403 (blocked)	Allows targeted backoff rather than treating all errors as generic failures
Deduplication	In-memory set tracking review IDs (`id`)	Ensures zero duplicated records when pulling across overlapping filters

3. Customer Satisfaction Drivers & Pain Points

A total of 179 unique reviews were harvested across pagination pages and sort filters (Lowest Rating, Highest Rating, Most Helpful). Text was cleaned and processed through VADER sentiment scoring and mapped into functional aspect tags:

Aspect	Positive Tag (Pos)	Negative Tag (Neg)	Customer Consensus

Audio Quality	Audio Quality & Sound Clarity (Pos)	Poor Sound Quality & Distortion (Neg)	Praised for clear vocals; criticized for hollow highs at loud volume.
Bass	Good Bass & Acoustic Performance (Pos)	Weak Bass & Flat Audio (Neg)	Decent bass response for the price; not deep enough for bass enthusiasts.
Comfort & Fit	Comfortable Fit & Lightweight (Pos)	Comfort & Ergonomics Concern (Neg)	Very lightweight on the ears; main complaint is lack of headband padding.
Price & Value	Great Value & Affordability (Pos)	Poor Value & Overpriced (Neg)	Consistently praised as one of the best budget wired headphones ($20–$25 CAD).
Build Quality	Good Design & Build Quality (Pos)	Durability & Build Concern (Neg)	Swivel fold-flat design is liked; thin cable is noted as the main durability weak point.
Recommendation	Satisfactory & Recommended (Pos)	Unsatisfactory & Not Recommended (Neg)	81% overall positive sentiment rate; widely recommended for students.

4. Business Recommendations

Area	Recommendation	Expected Impact
Product Listing	Clearly state 'No In-Line Microphone' in the product title and bullet points on BestBuy.ca.	Reduces returns from customers needing a headset for Zoom/work calls.
Product Improvement	Add basic rubber/silicone padding along the headband and reinforce the 3.5mm plug junction.	Directly addresses the two most common 1-star complaints (head soreness and broken wires).
Targeted Marketing	Position the product as an entry-level fleet choice for schools, testing centers, and libraries.	Opens reliable bulk B2B sales channels where low cost and durability are prioritized.